

ATWEEL TEA

PREMIUM QUALITY ASSURANCE

STANDARD OPERATING PROCEDURE (SOP)

End-to-End Quality Assurance, Production, and DDP Logistics Protocol

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Compliance Target	0.0% Adulteration 0.0% Moisture Gain 0% Pesticide Residue	Review Cycle	Annual
Certifications	FSSAI Certified ISO 22000:2018 HACCP Compliant GMP Audited		

CONFIDENTIALITY & COMPLIANCE NOTE: This document defines the international quality framework followed by ATWEEL TEA. Shared with global buyers to verify compliance with EU, US-FDA, and statutory global food safety norms.

1. OBJECTIVE & SCOPE

This document establishes absolute operational protocols across the entire value chain of **ATWEEL TEA**—spanning from sustainable plantation management, fine harvesting, chemical-free processing, clinical personnel hygiene, medical grade vacuum-packaging, climate-controlled warehousing, to unbreachable DDP (Delivered Duty Paid) ocean transit. The absolute target is achieving international award-grade sensory profiles while maintaining zero-tolerance for chemical residues and microbial defects.

2. SUSTAINABLE PLANTATION CARE & SOIL MANAGEMENT

Protocol for ecosystem balancing and soil nutrient management at the source gardens.

- CRITICAL Soil Analysis & pH Optimization:** Comprehensive soil testing must be executed annually. Soil pH levels must be strictly regulated and stabilized between **4.5 and 5.5**. Agricultural dolomite or lime applications must follow exact prescriptive agronomic charts.
- STANDARD Leguminous Shade Forestry:** Maintain systematic shade canopy ratios utilizing nitrogen-fixing species like *Albizia chinensis* and *Indigofera teysmannii*. Routine loop-sided pruning ensures filtered sunlight penetration, mitigating moisture stress on tea bushes.
- CRITICAL Non-Chemical Weed Management:** The deployment of chemical herbicides (including Glyphosate) is strictly banned. Weed containment is accomplished entirely via manual weeding cycles and extensive organic mulching using dry leaf matter and pruning remnants.

3. APPROVED ORGANIC NUTRITION POLICY (KHAD)

Mandatory matrix for plant health, restricting synthetic chemical fertilizers to ensure zero trace contaminants.

- **CRITICAL Vermicompost & Mature FYM:** Only fully decomposed Farmyard Manure (FYM) or nutrient-rich Vermicompost is approved, applied at a strict rate of 2 to 3 kg per bush annually during the dormant cycle.
- **STANDARD Botanical Oil Cakes:** Fine-powdered Neem Cake (De-oiled) and Castor Cake must be incorporated into the root zone to supply organic nitrogen and act as a natural subterranean nematocide.
- **STANDARD Bio-Fertilizer Inoculation:** Apply active bio-cultures including *Azotobacter* (nitrogen-fixing bacteria) and Phosphate Solubilizing Bacteria (PSB) post every structural pruning phase to boost natural microbial activity in the soil.

4. ZERO-RESIDUE PEST CONTROL & CROP PROTECTION

Strict compliance framework to pass 500+ pesticide residue screens for EU and US-FDA admission.

- **CRITICAL Botanical Extracts (Azadirachtin):** For primary infestations like Red Spider Mites, Thrips, and Tea Mosquito Bugs, only medical-grade ****Azadirachtin (Neem Oil formulated at 10,000 PPM)**** is approved. Dilution ratios must remain at 2-3 ml per liter of water.
- **STANDARD Microbial Biopesticides:** Looper caterpillar containment must utilize ***Bacillus thuringiensis*** (Bt formulation). Outbreaks of white-flies or sucking pests must be managed via entomopathogenic fungi sprays like ***Beauveria bassiana***.
- **CRITICAL Pre-Harvest Interval (PHI) Rule:** In extreme emergencies, only highly eco-friendly approved compounds can be applied, under a mandatory ****21 to 30 days absolute Pre-Harvest Interval****. Zero spraying is permitted within 30 days of plucking to ensure absolute zero trace contamination in the final leaf.

5. FINE HARVESTING & LEAF SELECTION STANDARDS

Field level harvesting parameters aimed at maximizing polyphenols and award-grade liquors.

- **CRITICAL Fine Plucking Mandate:** Harvest workflows enforce a strict **"Two Leaves and a Bud"** standard. Coarse, fibrous, mature, or damaged leaves are rejected at the bush level by trained field quality auditors.
- **STANDARD Monsoon Crop Exclusion:** Award-grade export batches must be harvested strictly during the Second Flush period (May-June). High-moisture monsoon flushes are automatically routed out of the premium export line.
- **CRITICAL Aerate Crates Over Sacks:** Traditional nylon or jute sacks are prohibited. Harvested leaves are placed loosely into food-grade, highly ventilated plastic crates to eliminate cell bruising and premature thermal fermentation.

6. PROCESSING FACILITY CLEANLINESS & CHEMICAL CONTROLS

Factory floor restrictions aimed at eradicating heavy metal ingress and moisture variability.

- **CRITICAL Zero-Chemical Factory Perimeter:** A strict 500-meter chemical-free buffer zone is enforced around the factory facility. No pesticide or chemical spraying is permitted within this zone.
- **STANDARD Food-Grade Stainless Steel Line:** All contact surfaces across Withering troughs, CTC rolling units, and Fermentation beds must consist entirely of certified Food-Grade **304/316 Stainless Steel**. Iron particle contamination tolerance is absolute zero.
- **CRITICAL Moisture Floor Target:** Factory drier discharge parameters must systematically lock moisture content between **3.0% and 3.5% maximum**, monitored via real-time online moisture analyzers.

7. PERSONNEL HYGIENE & HEALTH STANDARDS (GMP COMPLIANCE)

Sanitation parameters for factory operators and handling staff to avoid biological contamination.

- **CRITICAL** **Mandatory PPE Attire:** No worker is allowed onto the processing or packaging floor without complete sanitised factory uniform, high-coverage hairnets, nose masks, and food-grade nitrile gloves. Hairnets must wrap the ears entirely.
- **CRITICAL** **Multi-Stage Sanitization Station:** Handwashing stations with automated non-touch soap dispensers and medical alcohol rubs must be utilized at all entrances. Re-entry requires compulsory re-sanitization.
- **STANDARD** **Zero Personal Effects Policy:** Jewelry, watches, rings, and mobile phones are strictly barred from processing bays to completely eliminate any risk of accidental physical contamination.
- **CRITICAL** **Medical Clearance Protocols:** All staff undergo rigorous annual medical screening. Personnel showing active symptoms of respiratory illness, skin conditions, or open wounds are immediately repositioned to non-contact administrative roles.

8. ADVANCED 'SUPER-BARRIER' VACUUM PACKAGING

Engineering standards to preserve volatile flavor compounds and eliminate ambient air exposure.

- **CRITICAL Zero-WVTR Aluminum Bags:** Bulk and retail units must be packed into 4-Layer Aluminum Foil Barrier Liners possessing a Water Vapor Transmission Rate (WVTR) of 0.00. Liners must hold BPA-Free and EU EC No 10/2011 compliance certification.
- **STANDARD Double Nitrogen Gas Flushing:** Heavy vacuum machinery must extract 100% of residual ambient oxygen, replacing it with 99.99% pure, Bone-Dry Nitrogen Gas to permanently suspend oxidation processes for up to 24 months.
- **CRITICAL Puncture & Seal Integrity Audit:** Finished packs must pass automated high-sensitivity metal detection and inline pneumatic pressure tests to find any micro-punctures caused by sharp CTC tea grains.

9. CLIMATE-CONTROLLED STORAGE & TERMINAL ISOLATION

Intermediate port transit storage parameters at port terminal warehouses.

- **STANDARD Grade-A Dehumidified Warehousing:** At terminal storage locations (e.g., Kolkata Port), tea must be kept exclusively within Grade-A climate chambers equipped with 24/7 heavy-duty industrial dehumidifiers.
- **CRITICAL Environmental Metrics:** Terminal vault conditions must be digitally fixed at **20°C to 22°C temperature** and **50% to 55% Relative Humidity (RH)**.
- **CRITICAL Cross-Contamination Isolation:** Pallets must rest on food-grade plastic bases raised 6 inches off the concrete floor and placed 15 cm clear of walls. Storage alongside spices, volatile chemicals, or inside diesel-forklift pathways is totally prohibited.

10. PRE-SHIPMENT LABORATORY VERIFICATION

Final verification protocols before cargo loading approval.

- **CRITICAL Third-Party Independent Sampling:** Prior to container stuffing, certified scientists from international laboratories (SGS or Eurofins) must extract randomized composite samples directly from the sealed lot.
- **CRITICAL Comprehensive Safety Assays:** Samples are subjected to a full 500+ pesticide multi-complex screen, heavy metal testing (Lead, Arsenic), and mycotoxin assays (Ochratoxin A). Customs loading authorization is withheld until a 100% clean "Passed for EU/USA Food Entry" digital certificate is generated.

11. UNBREACHABLE CONTAINER PREPARATION & DDP TRANSIT

Logistics protective protocols spanning ocean shipping to destination warehouse delivery.

- **STANDARD Light & Odor Structural Audit:** Inspectors must enter the 20-foot container shell, close doors completely, and audit for light leaks. Any micro-fissure or residual smell of paint/petroleum triggers immediate container rejection.

- **CRITICAL Full Container Foil Liner:** The interior must be fitted with a full-size 20ft Aluminum Laminated Thermal Liner Bag. Tea pallets are stuffed inside this barrier shell to insulate the product from extreme marine temperature fluctuations.
- **STANDARD Desiccant Pole Matrix:** Suspend a minimum of 10 units of high-absorption Calcium Chloride (CaCl₂) desiccant poles along the side walls to eliminate cargo sweat risk.
- **STANDARD Reefer Option & DDP Execution:** For ultra-premium award lots, utilize Reefers programmed at a continuous +18°C. Under DDP compliance, all customs clearances, tariffs, and VAT duties are pre-cleared by our broker to guarantee immediate, unbroken delivery to the buyer's destination warehouse.

12. QUALITY METRICS & THRESHOLD SUMMARY MATRIX

OPERATIONAL PHASE	CRITICAL PARAMETER TARGET	FREQUENCY	VERIFYING AUTHORITY
Soil Health	pH 4.5 – 5.5 0% Heavy Metal Contamination	Annual Test	Agronomy Lab / QA Team
Crop Protection	Zero Synthetic Residues Neem Oil at 10,000 PPM	Per Application	Plantation Supervisor
Plucking Standard	100% Two Leaves and a Bud fine pluck	Every Batch / Hourly	Field Quality Auditor
Factory Output	Moisture: 3.0% – 3.5% 0% Ferrous Ingress	Hourly Digital Scan	Factory In-Charge
Packaging Liner	WVTR 0.00 Multi-Layer Aluminum Foil (BPA Free)	Per Lot Intake	Supplier Certification
Terminal Storage	Temp: 20°C–22°C Humidity: 50%–55% RH	Continuous Log	Automated Data Logger
Pre-Shipment Cert	0.0% Trace Detection across 500+ Chemicals	Per Export Lot	SGS / Eurofins Certified Report

13. TRACEABILITY & PRODUCT RECALL SYSTEM

ATWEEL TEA operates a rigorous backward-integration tracking infrastructure. Every export shipping unit is laser-coded with a unique multidimensional Batch Identifier. This barcode maps the finished tea product back to its precise blending timeframe, the processing shift machinery logs, the specific field section within the tea garden, and the comprehensive bio-input application history of that exact harvest window. Bi-annual mock recall simulations ensure our containment protocol can isolate any specific batch from international retail networks within 24 hours.

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